Exp No 4
1/8/26

Packet Sniffing Using Raw Socket in Python

Aim:-

To implement a packet sniffer using raw socket in python that captures live network traffic and parses Ethernet, IP, and transport-layer (TCP/UDP/ICMP) headers

Steps:-

Step 1: Setup the Lab Environment

Step 2: Install Visual Studio Code

Step 3: Verify python

Step 4: Create Project Folder

Step 5: Write the python Program

Step 6: Run Command prompt as Administrator

Step 7: Navigate and Run the Program.

Result:

Thus, the implementation of packet sniffer using raw socket in python executed successfully

```
import socket
import struct
sniffer = socket.socket(socket.AF_INET, socket.SOCK_RAW, socket.IPPROTO_IP)
host = socket.gethostbyname(socket.gethostname())
sniffer.bind((host, 0))
sniffer.setsockopt(socket.IPPROTO_IP, socket.IP_HDRINCL, 1)
sniffer.ioctl(socket.SIO_RCVALL, socket.RCVALL_ON)
print("Packet Sniffer Started...\n")
try:
    while True:
        raw_data, addr = sniffer.recvfrom(65535)
        ip_header = raw_data[0:20]
        iph = struct.unpack('!BBHHHBBH4s4s', ip_header)
        version = iph[0] >> 4
        ihl = iph[0] & 0xF
        ttl = iph[5]
        protocol = iph[6]
        source = socket.inet_ntoa(iph[8])
        destination = socket.inet_ntoa(iph[9])
        print("----------------------------------------")
        print("Version :", version)
        print("Header Length:", ihl * 4)
        print("TTL :", ttl)
        print("Protocol :", protocol)
        print("Source IP :", source)
        print("Destination :", destination)
        print("Packet Length:", len(raw_data))
except KeyboardInterrupt:
    sniffer.ioctl(socket.SIO_RCVALL, socket.RCVALL_OFF)
    print("\nStopped")
```

```
TERMINAL PORTS

ket Sniffer Started...

sion : 4
der length: 20
: 1
tocol : 17
rce IP : 192.168.56.1
tination : 224.0.0.251
ket length: 56

sion : 4
der length: 20
: 1
tocol : 17
rce IP : 192.168.56.1
tination : 224.0.0.251
```